

SCALABILITY & FUTURE PLAN

OptiCrop: Smart Agricultural Production Optimization Engine

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Team ID	SWTID-2026-7253
Team Size	5
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Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Marks

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future[cite: 46]. It also captures the team's roadmap for enhancing and evolving the project beyond its current state[cite: 46].

Evolutionary Feature Roadmap

Phase	Planned Feature	Target Timeline	Expected Impact
Phase 1 (Short-term: 1-6 mos)[cite: 46]	Automated Model Retraining & Hyperparameter Tuning Pipeline[cite: 46]	03 Jan 2027[cite: 46]	Maintained model performance over time and automated continuous optimization[cite: 46].
Phase 2 (Mid-term: 12 mos)[cite: 46]	Feature Enrichment (Soil Interactions and Geographic Profiles)	03 July 2027[cite: 46]	Improved model recommendation accuracy by 5-8% on edge case micro-climate variations.
Phase 3 (Long-term: 18+ mos)	Explainable AI (XAI) Dashboard for Stakeholders & Policymakers	03 Jan 2028[cite: 44, 46]	Increased transparency in recommendation decisions, explaining underlying biochemical factors.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Limited historical soil and crop dataset diversity range (e.g., only 2 years)[cite: 46].	Reduced model training dataset diversity and potential for temporal climate bias[cite: 46].	Medium[cite: 46]
2	High prediction latency during peak usage hours (single synchronous worker thread)[cite: 46].	Degraded user field experience and increased request timeouts on the Flask app[cite: 46].	High[cite: 46]
3	Basic feature engineering lacking in-depth soil-climate interaction profiles.	Sub-optimal model accuracy for certain localized soil regions and uncommon crop classes.	Medium[cite: 46]

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance, synchronous Flask application[cite: 46].	Implement Gunicorn with multi-workers and horizontal autoscaling using Docker on Render clusters[cite: 46].
2	Data Storage Performance	Single, non-replicated local MySQL instance[cite: 46].	Migrate to a managed MySQL cluster (e.g., on Render) with read-replicas for load balancing and high availability[cite: 46].
3	Model Inference	Model inference executes on the main application thread[cite: 46].	Offload inference requests to a separate, asynchronous task queue (Celery/Redis) and optimize model runtime with ONNX[cite: 46].
4	Security	Basic API key authentication with unencrypted storage[cite: 46].	Implement OAuth 2.0 with JWT, role-based access control, and full database-level encryption for sensitive metrics[cite: 46].

Future Roadmap

Phase / Area	Proposed Upgrade / Enhancement	Expected Performance Impact	Timeline
Phase 2[cite: 46]	- Migrate deployed model services to autoscaling cloud architecture (ECS with Fargate)[cite: 46]. - Database replication configuration (MySQL read-replicas)[cite: 46]. - Implement Content Delivery Network (CDN) for static frontend assets[cite: 46].	- Improved system responsiveness during peak usage seasonal loads[cite: 46]. - Higher concurrent user session support[cite: 46]. - Reduced global application layout latency[cite: 46].	Q1 2027[cite: 46]
Phase 3[cite: 46]	- Integrate continuous model monitoring and statistical drift detection APIs[cite: 46]. - Explore integration with advanced geographic anomaly detection frameworks[cite: 46]. - Leverage third-party global environmental datasets for enhanced feature engineering[cite: 46].	- Enhanced data availability and fault tolerance profiles[cite: 46]. - Proactive identification of model recommendation degradation[cite: 46]. - Potentially higher predictive output precision[cite: 46]. - Expanded engine spatial analysis capabilities[cite: 46].	Q3 2027[cite: 46]
Phase 4[cite: 46]	- Hardening the core architectural security and authorization layers[cite: 44, 46]. - Developing a secured, lightweight mobile field operator application interface[cite: 46]. - Expanding model applicability to new regional crop and soil seed sub-domains[cite: 46]. - Implement continuous CI/CD pipeline structures for automated code deployments[cite: 46].	- Drastically reduced vulnerability to remote cyber-attacks[cite: 46]. - Increased market outreach, usability, and rural field accessibility[cite: 46]. - Higher overall system automated reliability and runtime strategic values[cite: 46].	Q1 2028[cite: 46]